

BI Project

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IBM Watson Marketing Customer Value Data

Dataset: <https://www.kaggle.com/datasets/pankajjsh06/ibm-watson-marketing-customer-value-data>

Data Knowledge:

This dataset provides comprehensive information about customers of an auto insurance company, including various attributes that can be analyzed for strategic decision-making. It includes columns such as Customer Lifetime Value, which indicates the projected revenue from a customer over their relationship with the company, and Total Claim Amount, reflecting the financial impact of claims. Customer demographics are captured through fields like State, Education, Employment Status, Gender, Income, and Marital Status. Key policy-related details include Policy Type, Coverage, Number of Policies, and Monthly Premium Auto. Other important aspects are captured through metrics like Months Since Last Claim, Months Since Policy Inception, and Number of Open Complaints. The data also covers the effectiveness of different Sales Channels and Renew Offer Types. Additionally, Vehicle Class and Vehicle Size provide insights into the types of vehicles insured. This rich dataset is crucial for analyzing customer behavior, preferences, and the overall financial performance of the insurance company.

Business Problem:

The dashboard is designed to analyze customer behavior with the objective of improving customer retention strategies. By exploring demographics, response patterns, and customer lifetime value, the goal is to uncover actionable insights that drive targeted marketing, enhance sales strategies, and boost overall customer satisfaction. The focus is on leveraging data-driven decisions to minimize churn, maximize customer lifetime value, and optimize revenue generation.

Data Cleaning

```
In [3]: df.isnull().sum()
```

```
Out[3]: Customer          0
        State             0
        Customer Lifetime Value  0
        Response          0
        Coverage          0
        Education         0
        Effective To Date  0
        EmploymentStatus  0
        Gender            0
        Income            0
        Location Code     0
        Marital Status    0
        Monthly Premium Auto  0
        Months Since Last Claim  0
        Months Since Policy Inception  0
        Number of Open Complaints  0
        Number of Policies  0
        Policy Type       0
        Policy            0
        Renew Offer Type  0
        Sales Channel     0
        Total Claim Amount  0
        Vehicle Class     0
        Vehicle Size      0
        dtype: int64
```

No missing values

```
In [4]: duplicate_rows = df.duplicated()
        print("\nDuplicate rows:")
        print(duplicate_rows.sum())
```

```
Duplicate rows:
0
```

No duplicate Values

Distinct values of Customer:
['BU79786' 'QZ44356' 'AI49188' ... 'TD14365' 'UP19263' 'Y167826']

Distinct values of State:
['Washington' 'Arizona' 'Nevada' 'California' 'Oregon']

Distinct values of Response:
['No' 'Yes']

Distinct values of Coverage:
['Basic' 'Extended' 'Premium']

Distinct values of Education:
['Bachelor' 'College' 'Master' 'High School or Below' 'Doctor']

Distinct values of Effective To Date:
['2/24/11' '1/31/11' '2/19/11' '1/28/11' '2/3/11' '1/25/11' '1/18/11'
'1/26/11' '2/17/11' '2/21/11' '1/6/11' '2/6/11' '1/10/11' '1/17/11'
'1/5/11' '2/27/11' '1/14/11' '1/21/11' '2/5/11' '1/29/11' '2/28/11'
'2/12/11' '2/2/11' '2/7/11' '1/22/11' '2/13/11' '1/15/11' '1/8/11'
'1/11/11' '1/28/11' '2/8/11' '2/23/11' '1/2/11' '2/16/11' '1/27/11'
'1/23/11' '1/9/11' '2/11/11' '2/4/11' '2/1/11' '2/15/11' '2/26/11'
'1/16/11' '1/1/11' '2/10/11' '1/24/11' '2/25/11' '1/12/11' '2/9/11'
'1/19/11' '1/4/11' '2/14/11' '2/20/11' '2/18/11' '1/3/11' '1/13/11'
'1/30/11' '2/22/11' '1/7/11']

Distinct values of EmploymentStatus:
['Employed' 'Unemployed' 'Medical Leave' 'Disabled' 'Retired']

Distinct values of Gender:
['F' 'M']

Distinct values of Location Code:
['Suburban' 'Rural' 'Urban']

Distinct values of Marital Status:
['Married' 'Single' 'Divorced']

Distinct values of Policy Type:
['Corporate Auto' 'Personal Auto' 'Special Auto']

Distinct values of Policy:
['Corporate L3' 'Personal L3' 'Corporate L2' 'Personal L1' 'Special L2'
'Corporate L1' 'Personal L2' 'Special L1' 'Special L3']

Distinct values of Renew Offer Type:
['Offer1' 'Offer3' 'Offer2' 'Offer4']

Distinct values of Sales Channel:
['Agent' 'Call Center' 'Web' 'Branch']

Distinct values of Vehicle Class:
['Two-Door Car' 'Four-Door Car' 'SUV' 'Luxury SUV' 'Sports Car'
'Luxury Car']

Distinct values of Vehicle Size:
['Medsized' 'Small' 'Large']

No spelling mistakes and all the distincts values are correct

```

Data types:
Customer          object
State             object
Customer Lifetime Value  float64
Response          object
Coverage          object
Education         object
Effective To Date object
EmploymentStatus  object
Gender            object
Income            int64
Location Code     object
Marital Status    object
Monthly Premium Auto  int64
Months Since Last Claim  int64
Months Since Policy Inception  int64
Number of Open Complaints  int64
Number of Policies    int64
Policy Type       object
Policy            object
Renew Offer Type  object
Sales Channel     object
Total Claim Amount  float64
Vehicle Class     object
Vehicle Size      object
dtype: object

```

Correct data types

```

Maximum values:
Customer Lifetime Value      83325.381190
Income                       99981.000000
Monthly Premium Auto         298.000000
Months Since Last Claim      35.000000
Months Since Policy Inception 99.000000
Number of Open Complaints     5.000000
Number of Policies            9.000000
Total Claim Amount           2893.239678
dtype: float64

Minimum values:
Customer Lifetime Value      1898.007675
Income                       0.000000
Monthly Premium Auto         61.000000
Months Since Last Claim      0.000000
Months Since Policy Inception 0.000000
Number of Open Complaints     0.000000
Number of Policies            1.000000
Total Claim Amount           0.099007
dtype: float64

Average values:
Customer Lifetime Value      8004.940475
Income                       37657.380009
Monthly Premium Auto         93.219291
Months Since Last Claim      15.097000
Months Since Policy Inception 48.064594
Number of Open Complaints     0.384388
Number of Policies            2.966170
Total Claim Amount           434.088794
dtype: float64

```

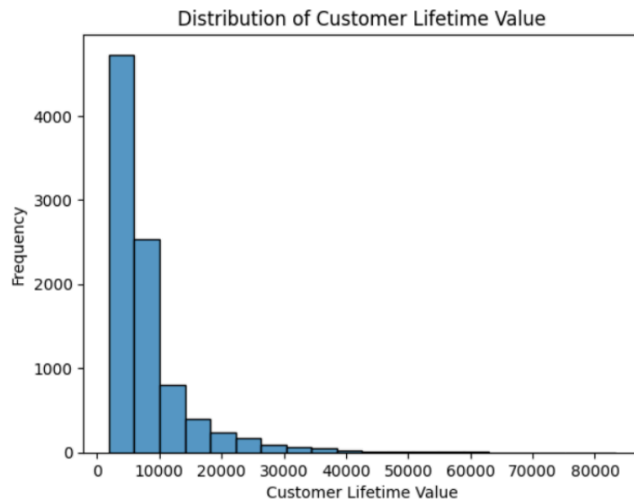
Showing values which dont have very high differences in their max and min

Conclusion: Data is clean and no need for data cleaning

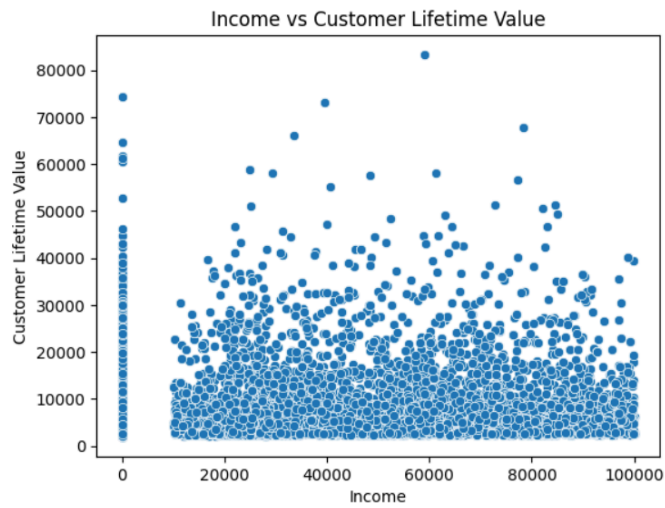
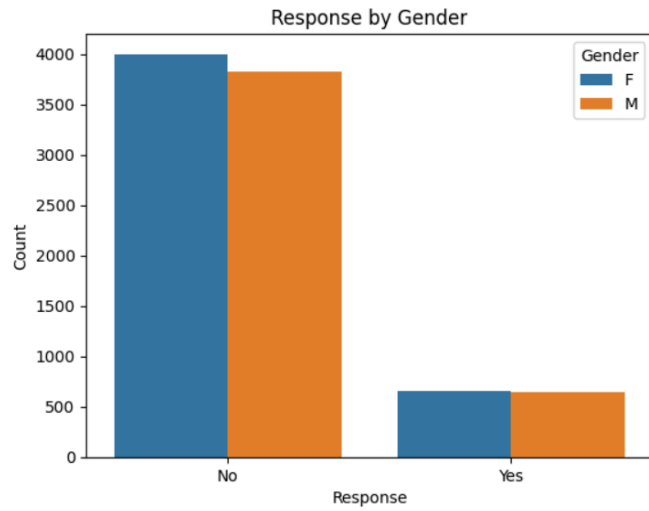
Data Wrangling and EDA

Frequency Distribution and Histograms:

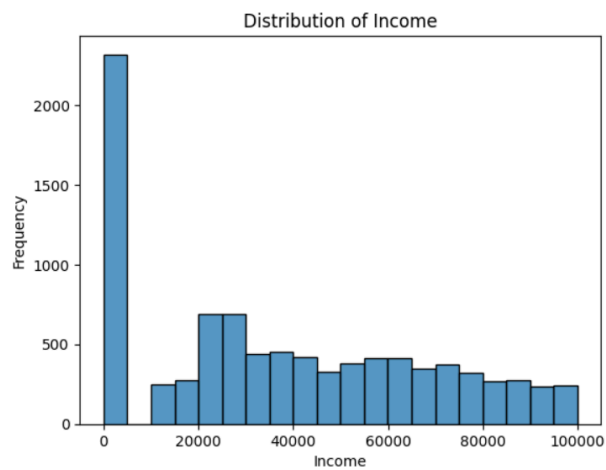
Using python codes, we constructed multiple frequency distribution plots, histograms, box plots, and scatter plots.



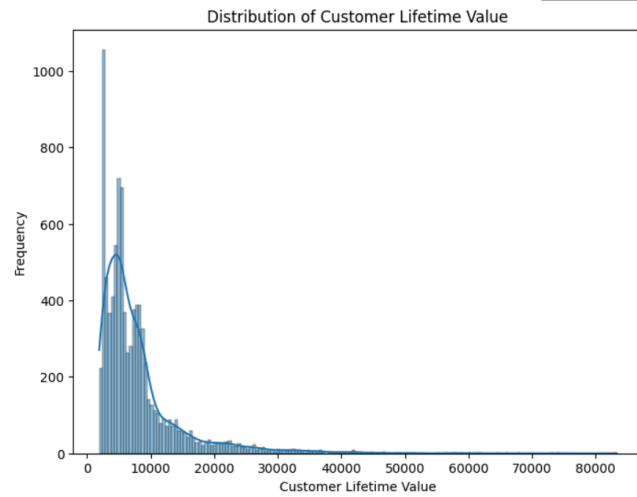
Customer Lifetime Value (CLV) value has a higher frequency at a lower rate.



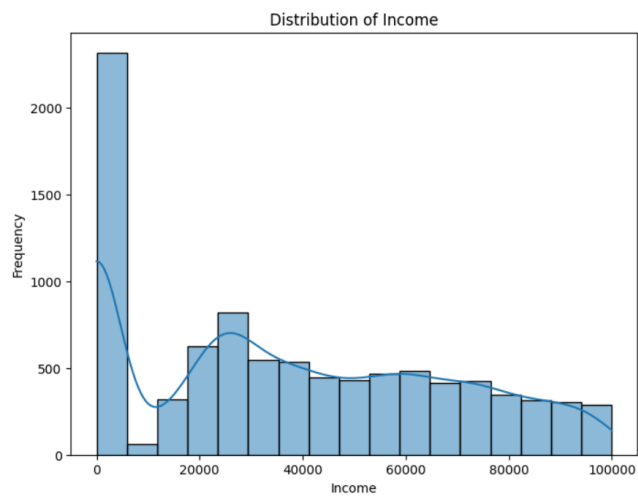
As CLV is increasing, income tends to decrease.



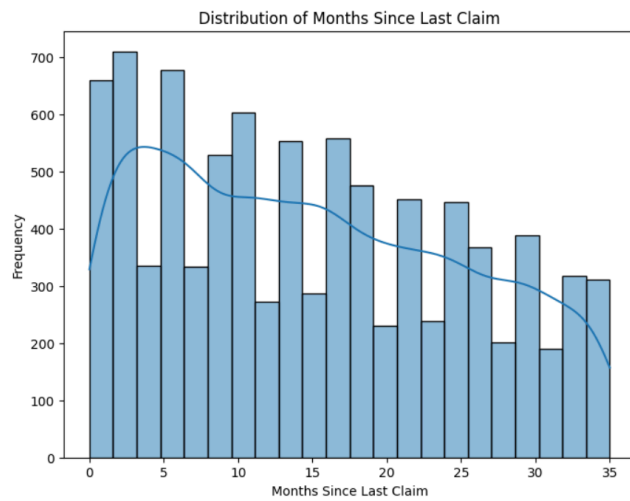
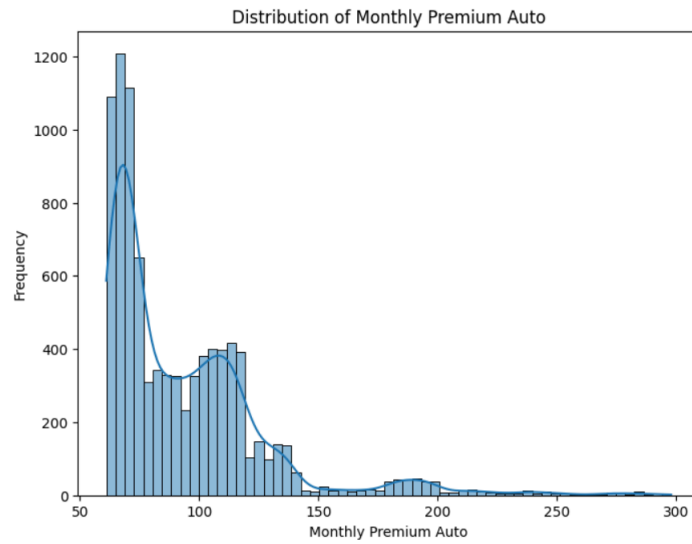
Highest frequency of income is 10000 with a frequency higher than 2000.



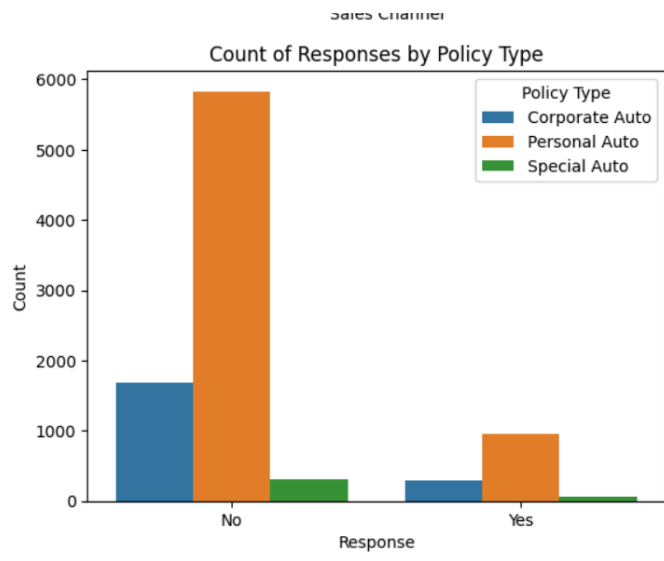
Positively **skewed** which means CLV is lower for most of the customers



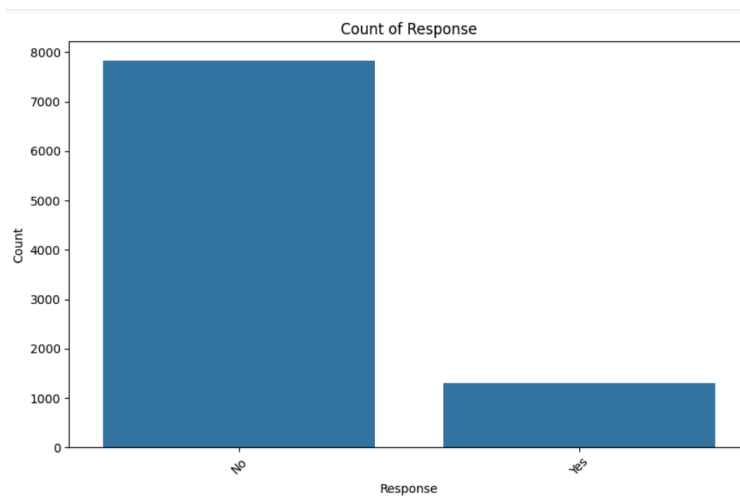
Same goes for income

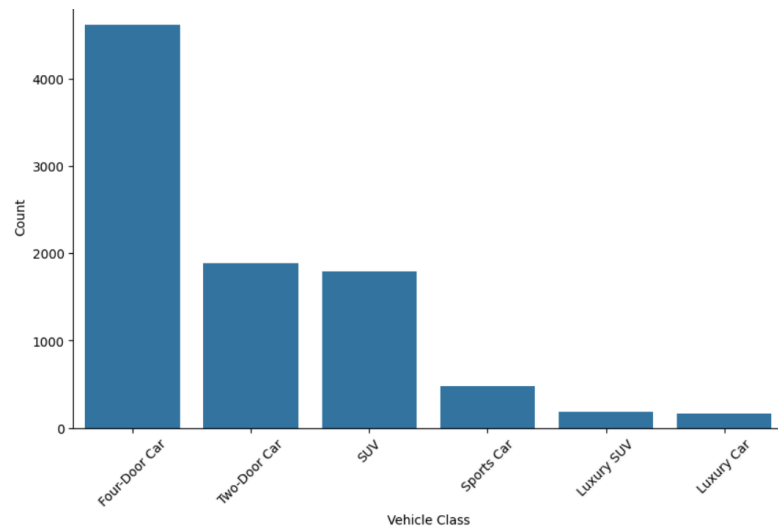


Hence from above distribution of numerical values, it can be interpreted that most of them are positively skewed and which are less for most of the customers

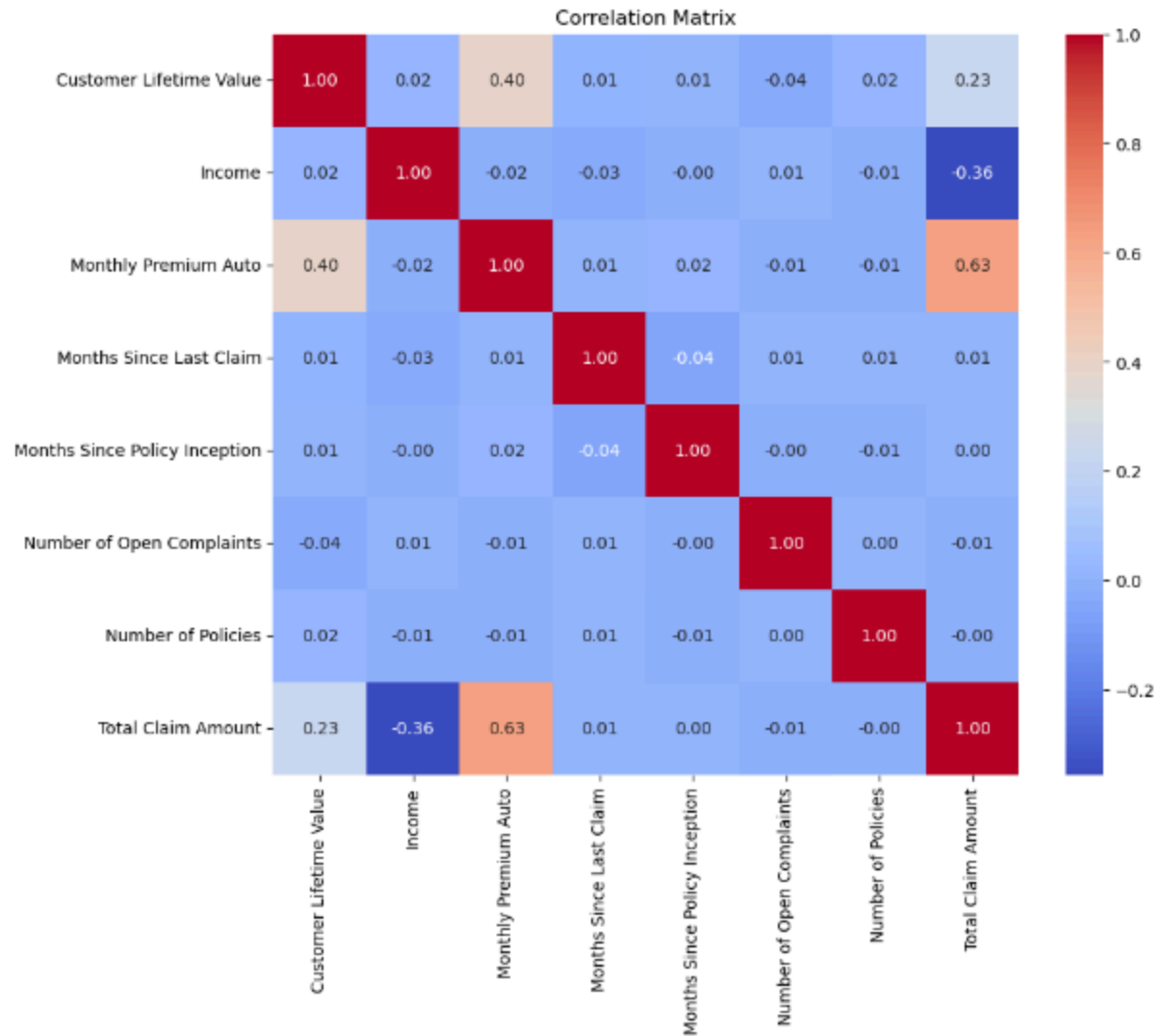


Most of the customers have negative response for the marketing campaign



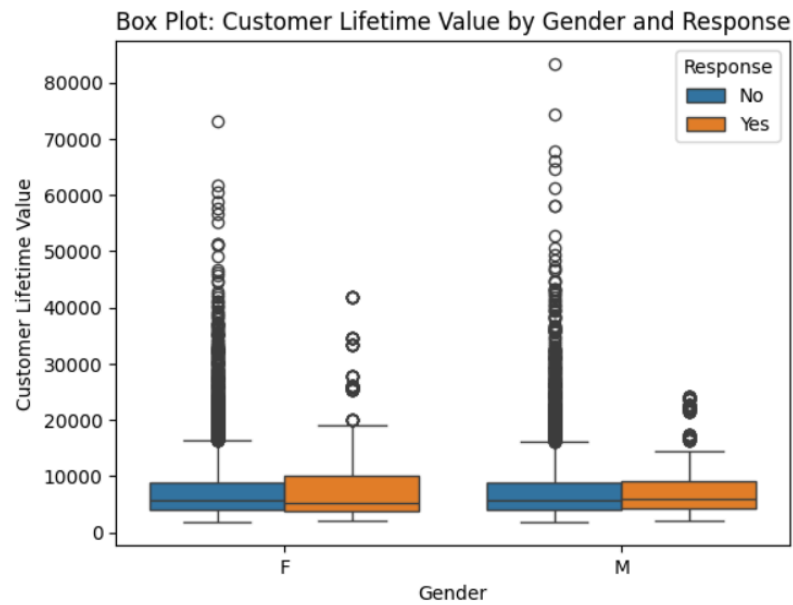


Most of the customers prefer Four Door car vehicle type

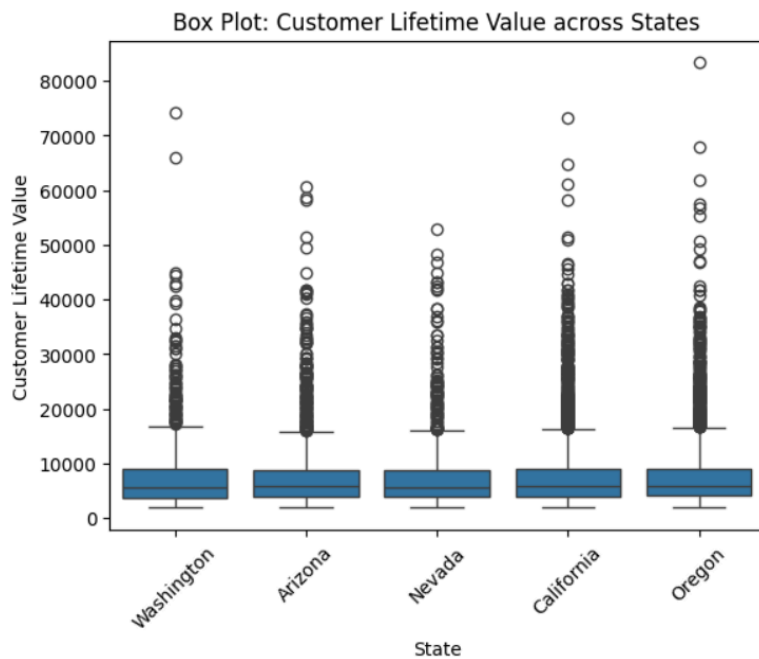


From this it can be observed that CLV is highly correlated with MPA and MPA is also highly correlated with TCM. While Income negatively correlated with TCM

Boxplots:

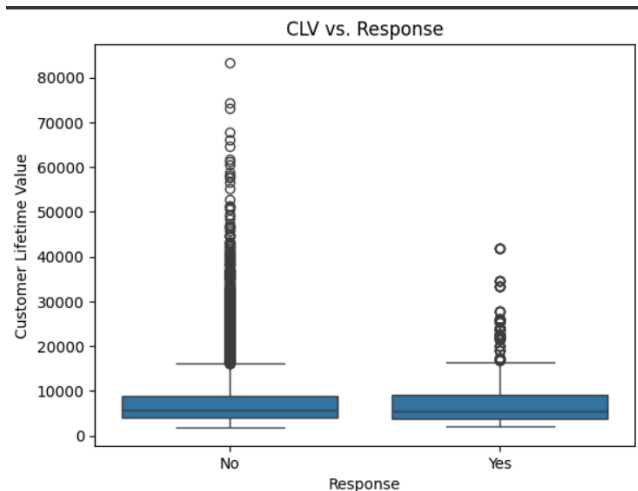
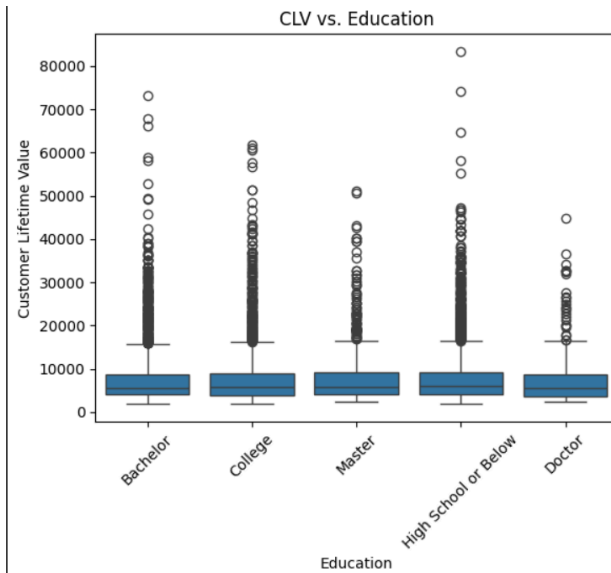


Not too much difference between male and female



Showing the CLV is not very much **different** among different places

Boxplots:



Chi-Square Test

Variable 1	Variable 2	Chi-Square	P-value	Significant
State	Response	0.438478	0.979207151	No
State	Coverage	3.25771	0.917159534	No
State	Education	16.60261	0.411750933	No
State	EmploymentStatus	4.783644	0.99672837	No
State	Gender	1.19525	0.878880018	No

State	Location Code	1.259353	0.996020147	No
State	Marital Status	9.432694	0.307118417	No
State	Policy Type	17.25113	0.027597715	Yes
State	Policy	67.14516	0.00027174	Yes
State	Renew Offer Type	2.515547	0.998104611	No
State	Sales Channel	8.416149	0.751823148	No
State	Vehicle Class	19.76305	0.47283732	No
State	Vehicle Size	9.750147	0.283002116	No
Response	Coverage	0.061277	0.969826324	No
Response	Education	10.97769	0.026815866	Yes
Response	EmploymentStatus	850.6926	8.02E-183	Yes
Response	Gender	0.152316	0.696331476	No
Response	Location Code	125.1301	6.74E-28	Yes
Response	Marital Status	117.5988	2.91E-26	Yes
Response	Policy Type	1.73063	0.420918978	No
Response	Policy	9.423022	0.307875709	No
Response	Renew Offer Type	548.1645	1.74E-118	Yes
Response	Sales Channel	107.4724	3.84E-23	Yes
Response	Vehicle Class	21.21024	0.000739213	Yes
Response	Vehicle Size	23.51373	7.84E-06	Yes
Coverage	Education	18.65114	0.016840896	Yes
Coverage	EmploymentStatus	21.76657	0.00536716	Yes
Coverage	Gender	2.830831	0.242824672	No
Coverage	Location Code	31.26975	2.70E-06	Yes
Coverage	Marital Status	1.591716	0.81028032	No

Coverage	Policy Type	1.272255	0.86606497	No
Coverage	Policy	16.72563	0.403562343	No
Coverage	Renew Offer Type	43.67369	8.58E-08	Yes
Coverage	Sales Channel	15.01889	0.020110277	Yes
Coverage	Vehicle Class	18.30847	0.049977868	Yes
Coverage	Vehicle Size	4.338917	0.362077211	No
Education	EmploymentStatus	148.7578	1.37E-23	Yes
Education	Gender	2.999092	0.557977302	No
Education	Location Code	291.6886	2.42E-58	Yes
Education	Marital Status	111.4186	1.95E-20	Yes
Education	Policy Type	7.769558	0.456298194	No
Education	Policy	25.04118	0.804357886	No
Education	Renew Offer Type	77.05513	1.50E-11	Yes
Education	Sales Channel	32.53495	0.001143582	Yes
Education	Vehicle Class	34.38951	0.023607445	Yes
Education	Vehicle Size	18.48837	0.017848772	Yes
EmploymentStatus	Gender	22.21613	0.000181511	Yes
EmploymentStatus	Location Code	1716.607	0	Yes
EmploymentStatus	Marital Status	1618.356	0	Yes
EmploymentStatus	Policy Type	1.959378	0.982231962	No
EmploymentStatus	Policy	43.69404	0.081468313	No
EmploymentStatus	Renew Offer Type	420.1676	2.02E-82	Yes
EmploymentStatus	Sales Channel	52.31394	5.45E-07	Yes
EmploymentStatus	Vehicle Class	28.34468	0.101487788	No
EmploymentStatus	Vehicle Size	20.61896	0.008231536	Yes

Gender	Location Code	68.83269	1.13E-15	Yes
Gender	Marital Status	17.43174	0.000163963	Yes
Gender	Policy Type	0.449661	0.798651533	No
Gender	Policy	2.84876	0.943487004	No
Gender	Renew Offer Type	22.48718	5.16E-05	Yes
Gender	Sales Channel	5.291652	0.151645024	No
Gender	Vehicle Class	8.71058	0.121180079	No
Gender	Vehicle Size	4.718767	0.094478431	No
Location Code	Marital Status	423.1334	2.79E-90	Yes
Location Code	Policy Type	2.901591	0.574426627	No
Location Code	Policy	6.52616	0.981350726	No
Location Code	Renew Offer Type	75.602	2.89E-14	Yes
Location Code	Sales Channel	2.498289	0.868659116	No
Location Code	Vehicle Class	45.47799	1.78E-06	Yes
Location Code	Vehicle Size	288.0044	4.19E-61	Yes
Marital Status	Policy Type	4.086774	0.394389586	No
Marital Status	Policy	19.00453	0.268428486	No
Marital Status	Renew Offer Type	290.8329	7.53E-60	Yes
Marital Status	Sales Channel	10.37677	0.109655889	No
Marital Status	Vehicle Class	29.51141	0.001029246	Yes
Marital Status	Vehicle Size	8.882944	0.064092848	No
Policy Type	Policy	18268	0	Yes
Policy Type	Renew Offer Type	4.83359	0.565327007	No
Policy Type	Sales Channel	4.655399	0.588707013	No
Policy Type	Vehicle Class	6.054829	0.810635907	No

Policy Type	Vehicle Size	1.474333	0.831178172	No
Policy	Renew Offer Type	21.38863	0.615706687	No
Policy	Sales Channel	19.59267	0.71968865	No
Policy	Vehicle Class	35.60937	0.668134835	No
Policy	Vehicle Size	10.12279	0.860137484	No
Renew Offer Type	Sales Channel	238.4739	2.70E-46	Yes
Renew Offer Type	Vehicle Class	120.054	1.85E-18	Yes
Renew Offer Type	Vehicle Size	6.482411	0.371370634	No
Sales Channel	Vehicle Class	34.36295	0.003027417	Yes
Sales Channel	Vehicle Size	7.788101	0.254042272	No
Vehicle Class	Vehicle Size	24.66172	0.006024951	Yes

Around 40% of the chi-squared tests appear to have a significant result, rest are not as significant.

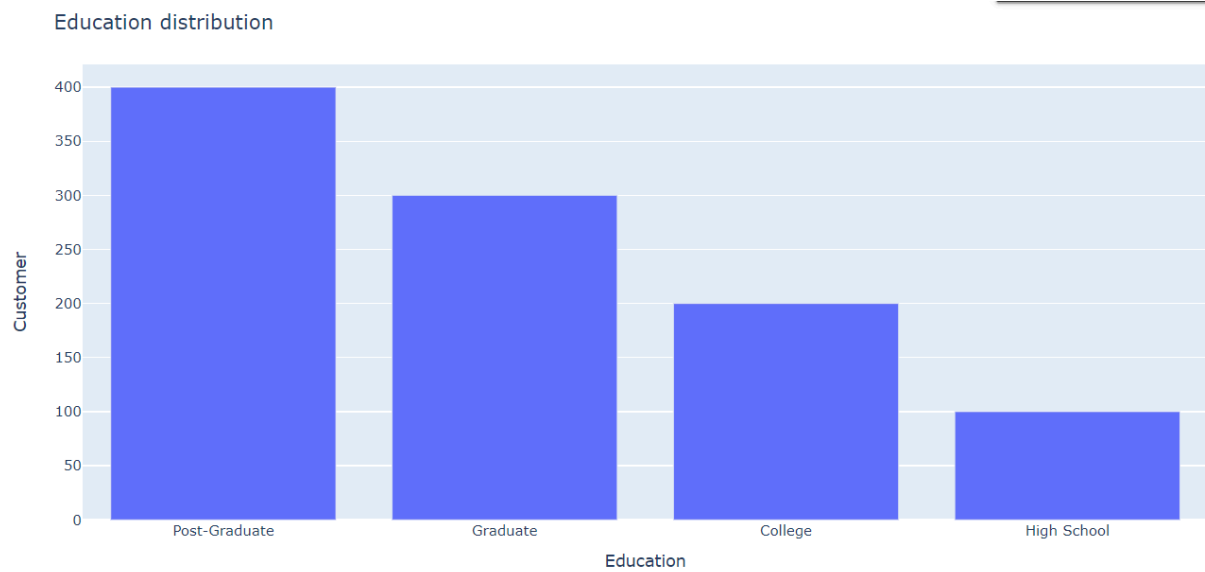
ANOVA TESTS:

Against customer lifetime value, we did a couple of tests. So, we got the following results:

	sum_sq	df	F	PR(>F)
C(State)	4.362581e+07	4.0	0.238113	9.169097e-01
C(Response)	2.242355e+07	1.0	0.489559	4.841421e-01
C(Coverage)	1.214361e+10	2.0	132.561742	1.782688e-57
C(Education)	3.802909e+08	4.0	2.075661	8.119421e-02
C(EmploymentStatus)	7.064115e+08	4.0	3.855655	3.919998e-03
C(Gender)	4.762320e+07	1.0	1.039726	3.079122e-01
C(Policy)	4.925742e+08	8.0	1.344256	2.161962e-01
Residual	4.172250e+11	9109.0	NaN	NaN

C(State) test is not significant
 C(Response) test is not significant
 C(Coverage) test is significant
 C(Education) test is not significant
 C(EmploymentStatus) test is significant
 C(Gender) test is not significant
 C(Policy) test is not significant
 Residual test is not significant

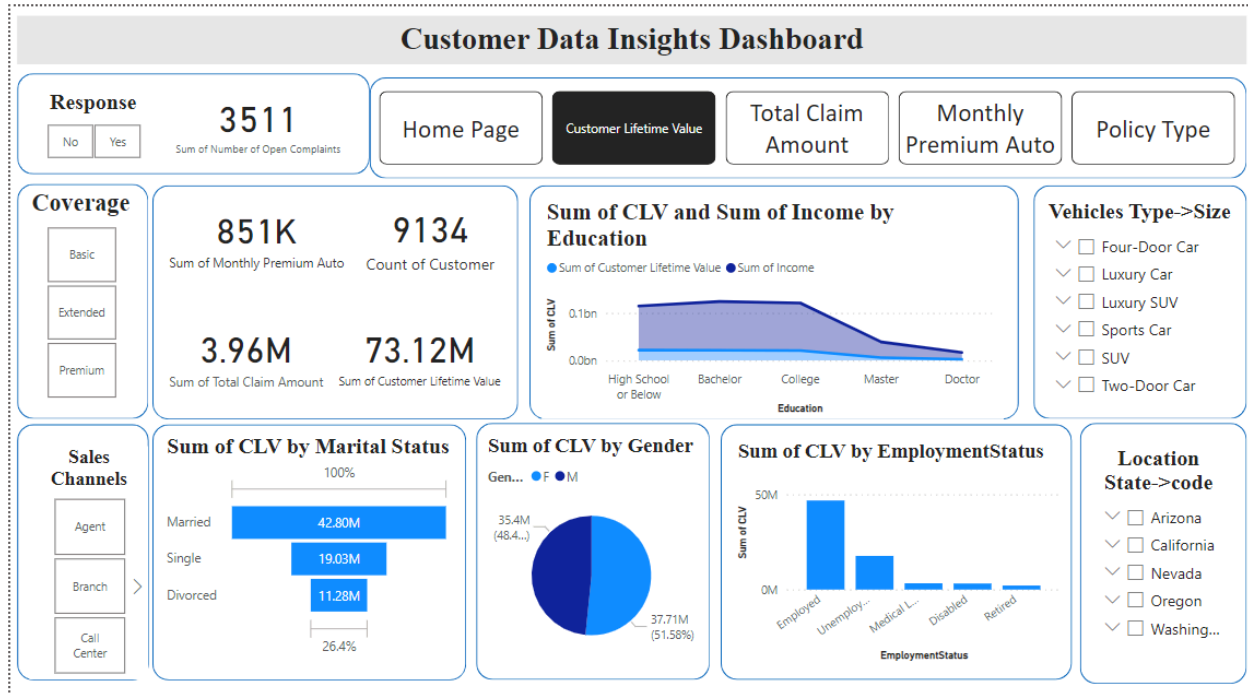
None of the tests were significant, so no need to perform Tukey either.



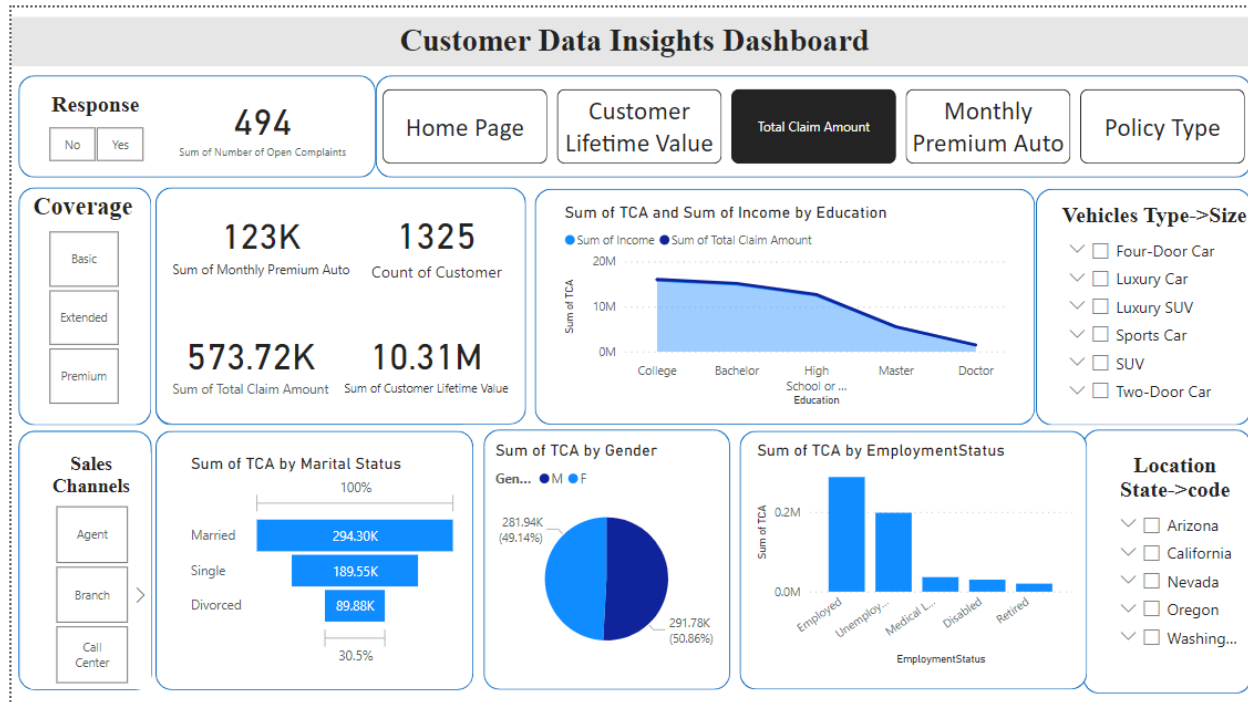
Most of the customers have post graduate level of education



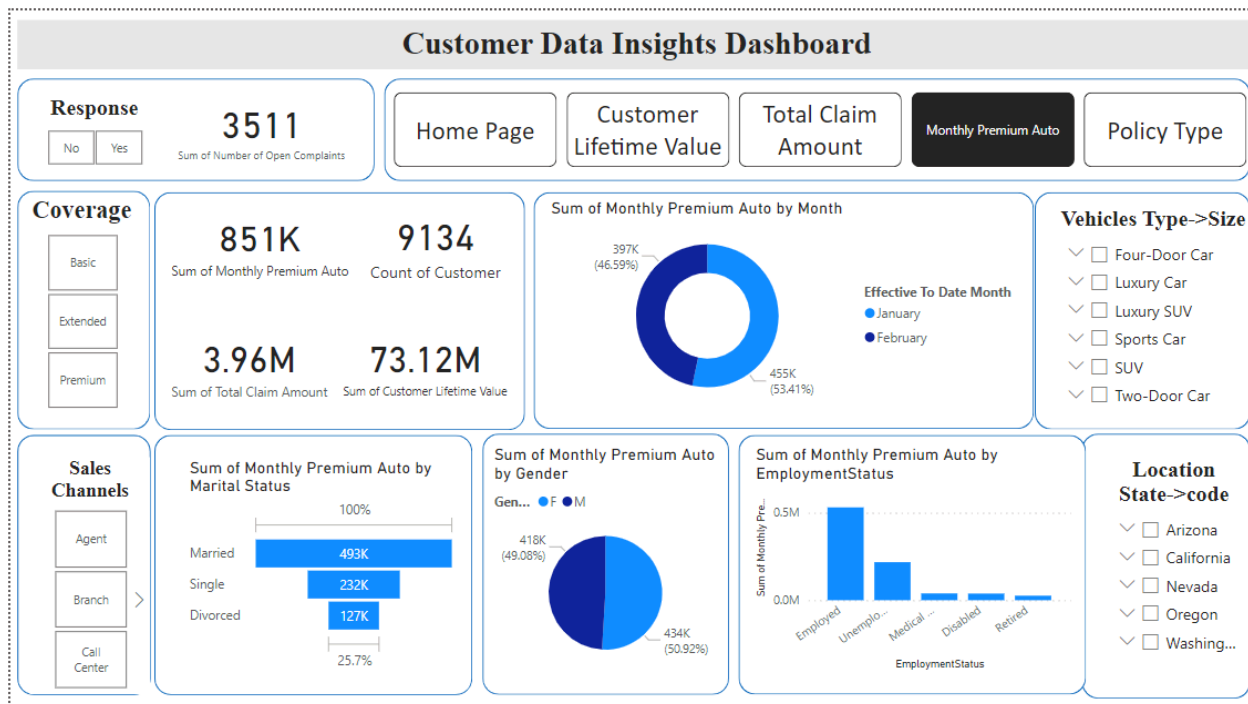
The homepage of the ALLState Auto Insurance dashboard serves as a gateway to unlocking valuable customer insights through the power of data analytics. This report provides an overview of the key features and functionalities of the homepage, highlighting its role in optimizing marketing strategies, enhancing sales approaches, maximizing customer satisfaction, and ultimately driving revenue growth for ALLState.



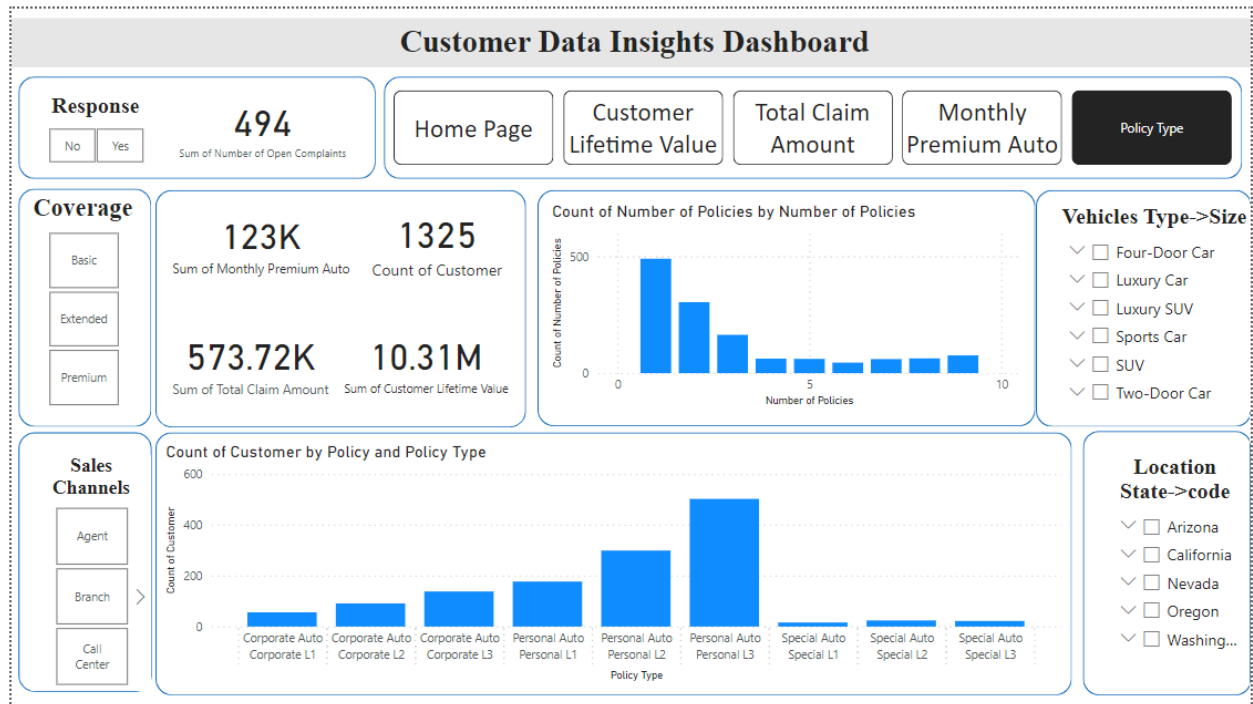
Welcome to the ALLState Auto Insurance Dashboard, your gateway to understanding revenue drivers and customer preferences. Divided into four essential sections - Customer Lifetime Value (CLV), Total Claim Amount, Monthly Premium Auto, and Policy Type - this dashboard centers on analyzing the company's revenue based on customer preferences. With a keen focus on customer behavior change and preferences, interactive slicers on the sides allow for a detailed examination of how different factors such as policy type, coverage, and vehicle choices influence revenue and customer demographics. Gain actionable insights to drive strategic decisions and enhance customer satisfaction.



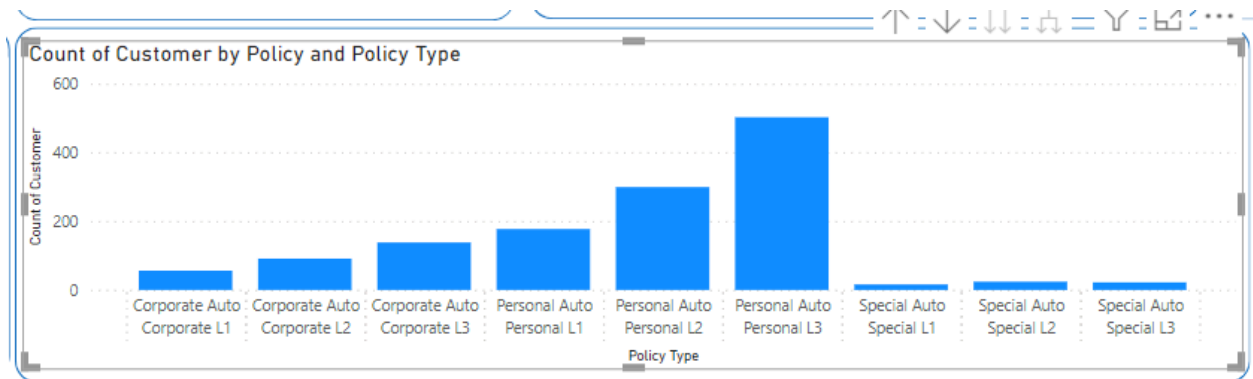
This page is for showing the Total Claim Amount in relation to customer behavior changes.



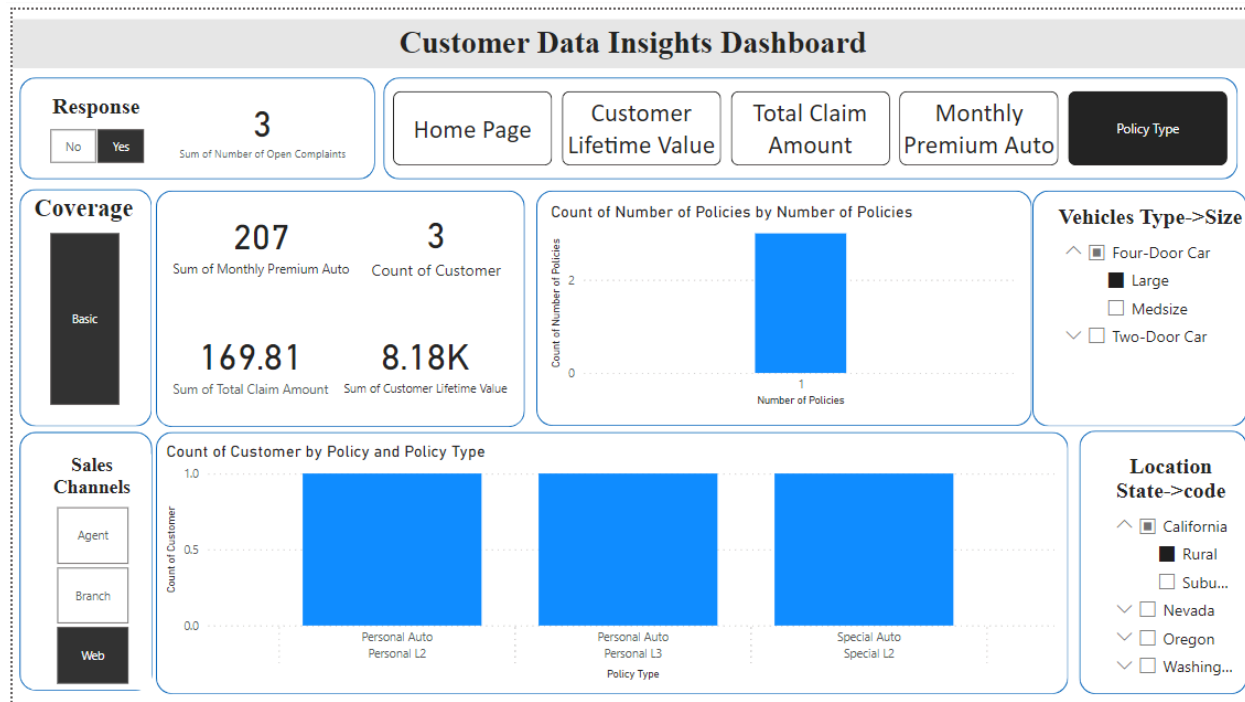
This page is for showing MPA relation with customers behavior change



This page is for showing Policy with customers' behavior relation



The feature of drill down is also added here to see the preference of customer for different policy type and policy



Here it is shown how the slicers work and how the visuals change for each slicer as shown in picture

Conclusion:

In conclusion, this dashboard effectively addresses our business problem statement by providing a comprehensive analysis of customer behavior aimed at enhancing retention strategies. By examining demographics, response patterns, and customer lifetime value, the dashboard uncovers actionable insights that inform targeted marketing efforts, refine sales approaches, and improve overall customer satisfaction. Through data-driven decision-making, it enables us to minimize churn, maximize customer lifetime value, and optimize revenue generation, thereby fulfilling our strategic objectives.